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import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import odeint

def main():

    problem_four("a")

    problem_four("b")

def problem_four(part):
    # Constants
    R = 5
    g = 9.8

    # Pendulum ODE system
    def pendulum(y, t):
        phi, omega = y

        dphi_dt = omega
        domega_dt = -(g / R) * np.sin(phi)

        return [dphi_dt, domega_dt]

    # Initial conditions
    if part.lower() == "a":
        phi0 = np.deg2rad(15)
        omega0 = 0

    elif part.lower() == "b":
        phi0 = np.deg2rad(90)
        omega0 = 0

    else:
        raise ValueError("Part must be 'a' or 'b'.")

    y0 = [phi0, omega0]

    # Time interval
    dt = 0.25
    tspan = np.arange(0, 10 + dt, dt)

    # Numerical solution
    sol = odeint(pendulum, y0, tspan)

    phi = sol[:, 0]

    # Small-angle approximation

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phi_small = phi0 * np.cos(np.sqrt(g / R) * tspan)

# Plot
plt.figure(figsize=(8, 5))

plt.plot(
    tspan,
    phi,
    label="Numerical solution",
    color="navy"
)

plt.plot(
    tspan,
    phi_small,
    "--",
    label="Small-angle approximation",
    color="red"
)

plt.xlabel("Time [s]")
plt.ylabel(r"$\phi$ [rad]")
plt.title(f"Pendulum Motion - Case {part.upper()}")

plt.grid(ls="--")
plt.legend()

plt.show()

# Run problem 4
if __name__ == "__main__":
    main()

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